

A Appendix

A.1 First online appendix: Ph.D. completion description table

Table A1. Ph.D. research detailed roadmap, trajectory, and timeline.

Stage	Phase	Tasks	Due date
Stage I	Problem identification and definition— RQ1	Literature review on DC adoption and EE	September 2024
		Two group-based EE with DC experts (user studies and data analysis)	February 2025
		Ph.D. proposal writing and defense	July 2025
		Manuscript (on user studies with DC experts)	October 2025
Stage II	Problem understanding— RQ2	LLMs (zero-shot) versus DC experts	September 2025
		LLMs (context-aware versus zero-shot)	October 2025
		LLMs (anchored versus context-aware)	November 2025
		Comparative studies—LLMs with DC experts: results analysis, and manuscript writing	February 2026
		Expertise-mix experiment: results analysis, and manuscript writing	May 2026
		Context synthesis experiment: results analysis, and manuscript writing	June 2026
Stage III	Prototype development and testing— RQ3 , and RQ4	Open-ended experiment: compilation, user study (DC experts raters), results analysis, and manuscript writing	September 2026
		Literature review on group EE: judgments collection and aggregation	August 2026
		Prototype implementation: medium fidelity	October 2026
		Pilot & 1 st user testing: usability and contextual inquiry	November 2026
		Prototype refinement	January 2027
		2 nd user testing: usability, user experience, and needs-assessment on the technology and the task	March 2027
		Manuscript on 2 nd user testing results	April 2027
		Comparative assessment: synchronous and asynchronous group EE	May 2027
Stage IV	Ph.D. completion	Manuscript on comparative assessment results	June 2027
		Refinement and adaptation Thesis writing and defense	June 2027 September 2027

A.2 The research questions

This section presents the research questions guiding this Ph.D. project.

- RQ₁** How do energy experts perceive uncertainties related to the adoption and use of Medium-Voltage (MV) Direct Current (DC) and Low-Voltage (LV) DC technologies?
- RQ₂** How does the use of virtual experts address the scarcity of energy experts, and ensure a mixture of expertise for participation in the Expert Elicitation (EE) studies?
- RQ₃** How does asynchronous participation in group-based EE study address human experts' time constraints?
- RQ₄** How do online group-based EE study outcomes differ when conducted with only human experts and when conducted with a mixture of human and virtual experts?